

Classification

Attendance



linktr.ee/CODE.CRUNCH



Supervised Learning

What is Supervised Learning?

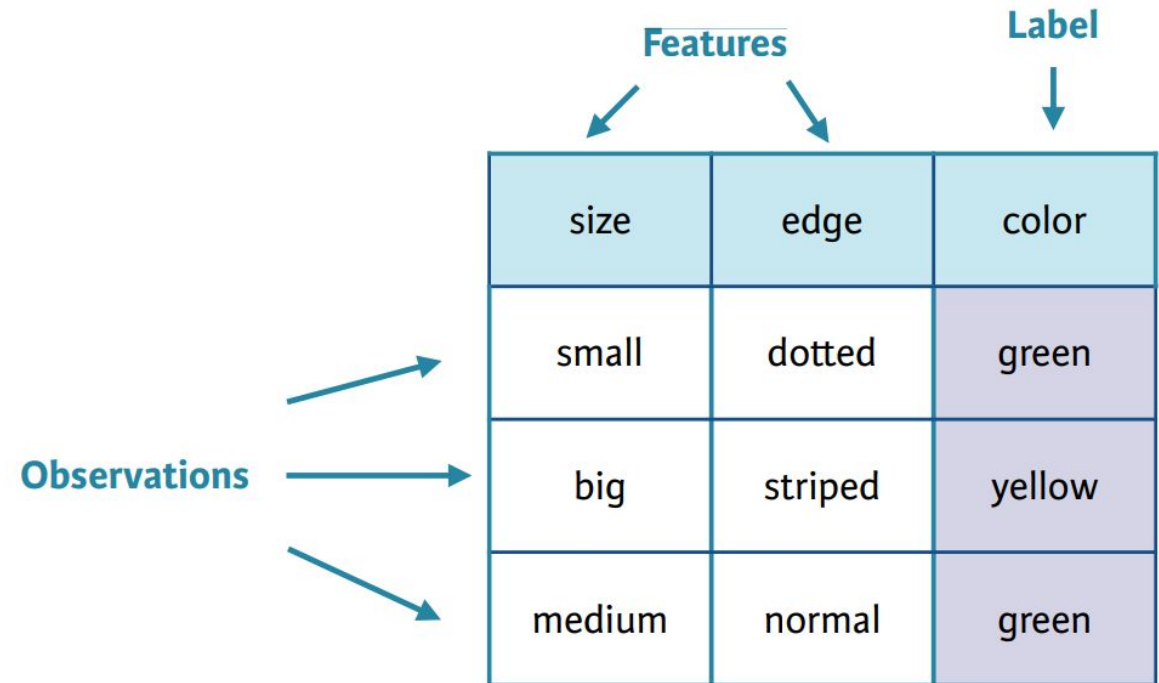
- Supervised learning is training a model to predict a labeled output feature based on input features
 - e.g., dog breed, car/truck, dog/cat, ...

When is an instance labeled?

- an instance is labeled if the output feature's value is known for that instance
 - e.g., instances have a distinction for dog breed, dog/cat, car/truck, ...

why?

- The purpose is to predict the output feature value for unlabeled instances
 - e.g, instances that do not have the distinction for dog breed, dog/cat, car/truck



	Features		Label
	size	edge	color
Observations	small	dotted	green
	big	striped	yellow
	medium	normal	green

Classification



What is Classification?

- a type of supervised learning task where the goal is to predict a categorical label or output feature

Common classification techniques:

- logistic regression
- decision trees
- randomforest
- support vector machines
- neural networks

The diagram illustrates a classification dataset using a table. The table has three columns: 'size', 'edge', and 'color'. The first two columns are grouped under the label 'Features' with arrows pointing to them. The third column is labeled 'Label' with an arrow pointing to it. There are three rows of data, each preceded by an arrow from the label 'Observations'. The data rows are: (small, dotted, green), (big, striped, yellow), and (medium, normal, green). The 'color' column is shaded light purple.

Features		Label
size	edge	color
small	dotted	green
big	striped	yellow
medium	normal	green

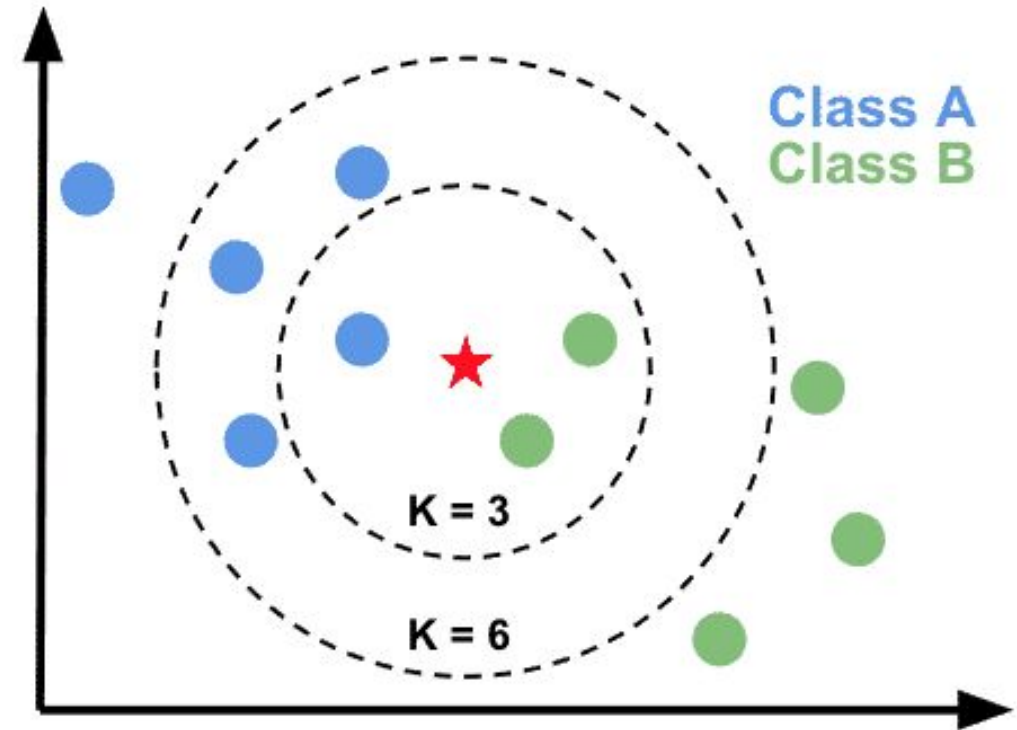
K - Nearest Neighbors

What is K-Nearest Neighbors?

- K-Nearest Neighbors (k-NN) is a simple yet powerful supervised learning algorithm
- Compute the distance between the new data point and all the points in the training dataset.
- It is based on the idea that similar data points tend to exist near each other in feature space.

Hyperparameters:

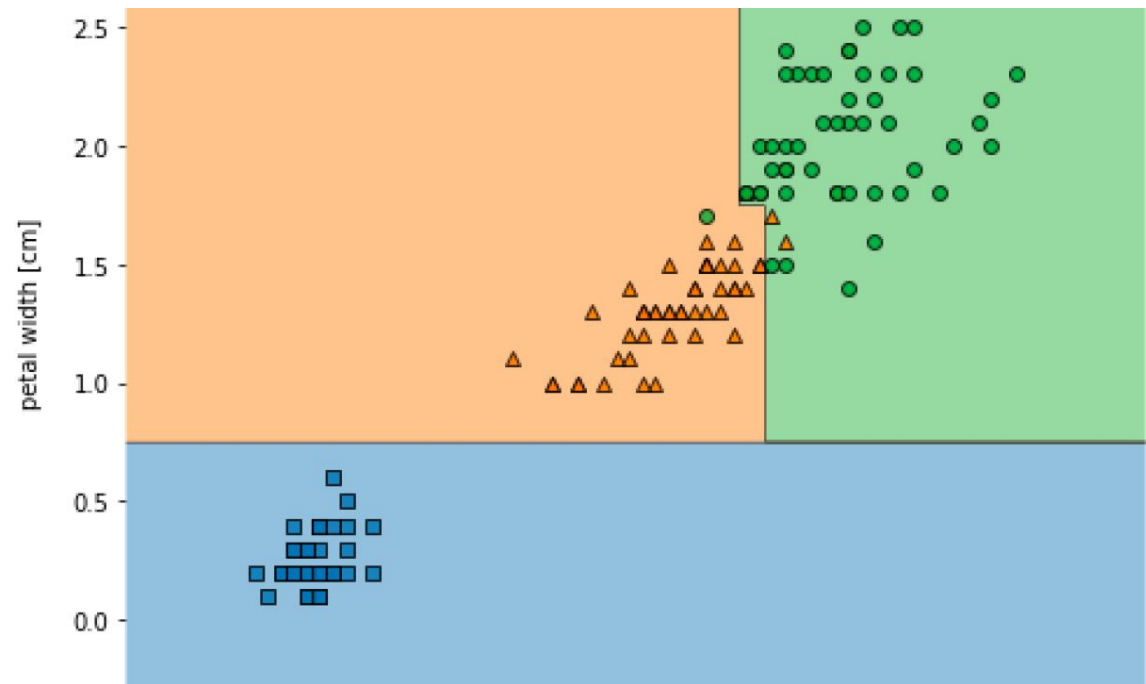
- K
 - Determines how many neighbors are considered when making predictions.
- Distance metric
 - Determines how similarity is measured between data points.



Decision Boundaries

What is a Decision boundary?

- represents the dividing line between predicting one class vs another class
 - useful for comparing models and identifying model weakness and strengths



logistic regression

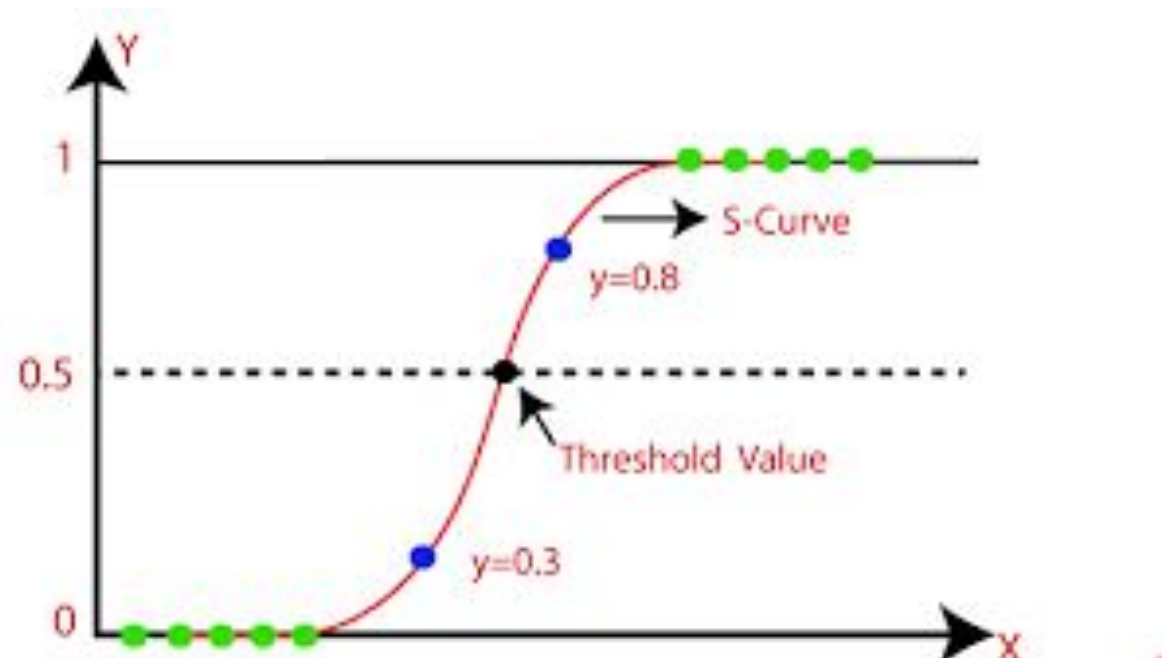
What is Logistic Regression?

- It predicts the probability of a data point belonging to a specific class by modeling the relationship between input features and the output as a logistic function

Hyper Parameters:

- Threshold
 - determines the class threshold

$$f(x) = w_1x_1 + \dots + w_nx_n$$



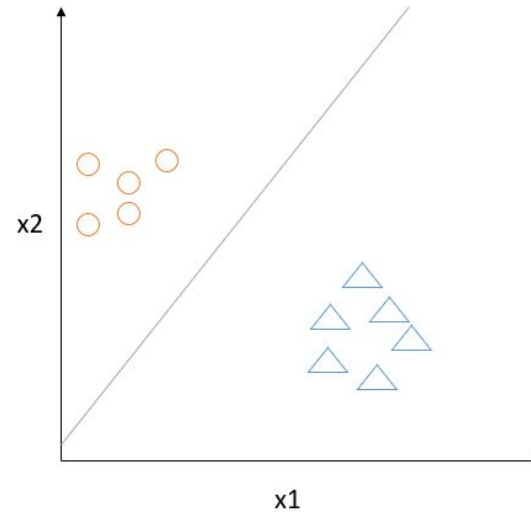
multi classification

What is multi-feature classification

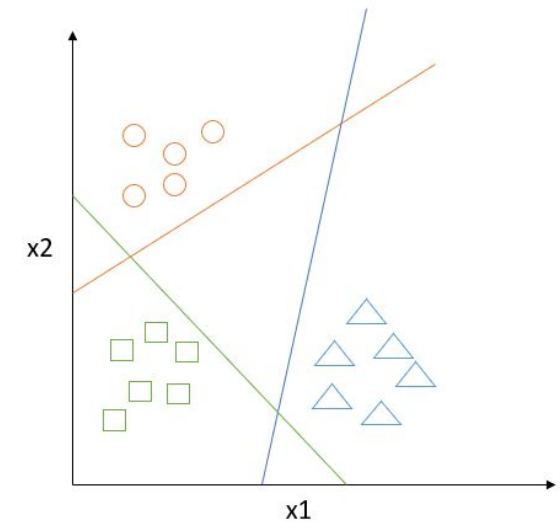
- classification tasks where the input consists of multiple features (or variables) that are used to predict a target class or label

Common Technique:

- One vs. All
 - classify one feature against all the others



Binomial Classification



One vs All Classification

Evaluating Classification models

accuracy

- Measures the percentage of correctly classified instances

Precision

- Measures the proportion of true positives among all predicted positives

Recall (Sensitivity)

- Measures the proportion of actual positives correctly identified

f1-score

- The harmonic mean of precision and recall

Specificity

- Measures the proportion of actual negatives correctly identified



Q&A Session



Attendance



linktr.ee/CODE.CRUNCH



Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



RSVP lu.ma/CODECRUNCH



Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch